

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is a lathe machine. Explain its working principle. Draw a neat labelled diagram of a lathe machine, showing its main parts.
- Q.37 Describe the main parts of a column and knee type milling machine and their functions.
- Q.38 What safety precautions should be taken while operating Drilling, Boring, shaping, Slotting and Planning machines?

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- A single cutter
- Simultaneous use of several cutters mounted on arbor

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- c) Engraving
- d) Planing action
- Q.6 CNC milling machine is controlled by:
 - a) Manual levers
 - b) Numeric code and programs
 - c) Gears only
 - d) Hand wheels
- Q.7 The cutting edges of a twist drill are located on the
 - a) Shank
 - b) Flutes
 - c) Lips
 - d) Web
- Q.8 The vertical boring machine is also called
 - a) Surface grinder
 - b) Turret lathe
 - c) Vertical turret lathe
 - d) Radial drilling machine
- Q.9 Shaping and planing require:
 - a) Low speeds and higher feeds for roughing
 - b) Always high speed
 - c) Multi point cutters
 - d) Use of coolant always
- Q.10 Lubricants are mainly designed to :
 - a) Cool the cutting area
 - b) Reduce friction between machine parts
 - c) Flush chips
 - d) Harden the tool

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name two common types of single point cutting tools.
- Q.12 Define a lathe chuck.

- Q.13 What is step turning?
- Q.14 What is an angle plate in lathe accessories?
- Q.15 The main principle of milling involves feeding the tool against a rotating cutter. (True/False)
- Q.16 The power required for milling is usually Higher than drilling for similar material removal. (True/False)
- Q.17 The rotating part that transmits power to the drill bit is called the _____.
- Q.18 A slotter machine differs from a shaper machine by reciprocating in a _____ direction.
- Q.19 The "Quick return" mechanism provides a rapid _____ stroke with no cutting.
- Q.20 The method of lubrication that involves manually applying oil is called _____ lubrication.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Compare the British and American systems for tool signature and explain their differences.
- Q.22 What is a tool signature? Explain with an neat sketch.
- Q.23 What specifications are given while ordering a lathe?
- Q.24 Distinguish between plain turning and step turning.
- Q.25 Explain the use of three-jaw, four-jaw and magnetic chucks.
- Q.26 When is CNC lathe technology preferred over conventional methods and why?
- Q.27 How does a horizontal milling machine differ from a vertical milling machine?